

Liam Sanchez

Embedded Systems & IoT Firmware Engineer • 6 Years Experience • MERIQ Index: 80% (Strong Candidate)

■ liam.sanchez724@example.com | ■ +1 (712) 731-3923 | ■ London, UK

TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: C/C++, I2C/SPI/UART, PCB Debugging, FreeRTOS, Python, Embedded Linux, ARM Cortex, MQTT

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Texas Instruments* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core C/C++ service, reducing latency by 38%.
- Led a squad of 5 engineers delivering scalable microservices processing 17M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 63%.

Associate Embedded Systems & IoT Firmware Engineer — *Bosch* (2020 - 2023)

- Designed and maintained I2C/SPI/UART data endpoints with 97% SLA uptime.
- Optimized database indexing and caching layers using PCB Debugging saving \$30k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [C/C++, I2C/SPI/UART, PCB Debugging, FreeRTOS]

Engineered end-to-end production solution leveraging C/C++ and I2C/SPI/UART with full test automation.

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [PCB Debugging, FreeRTOS, Python, Embedded Linux]

Architected high-resilience system integrating PCB Debugging with 99.9% uptime.

EDUCATION & CREDENTIALS

M.S. in Software Engineering

University of Toronto — 2016 - 2020 (Score: 3.77 / 4.0)